

CHAPTER 1. FUNCTIONS AND GRAPHS

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CONTENTS

- 1 What is Calculus?
- 2 Preliminaries
- 3 Lines in the plane; Parametric equations
- 4 Functions and Graphs
- 5 Inverse functions

1.4 Functions and Graphs

Definition 1 (Function)

A **function** f is a rule that assigns to each element $x \in D$ exactly one element y , denoted by $f(x)$ the value of f at x and is read “ f of x ”.



- The set D is called the **domain** of the function.
- The **range** of f is the set of all possible values of $f(x)$ as x varies throughout the domain, i.e.,

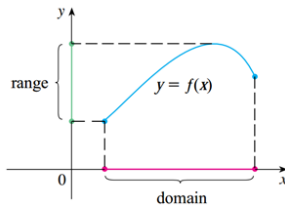
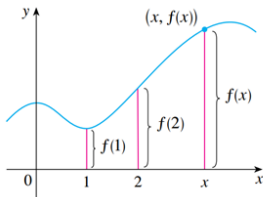
$$R = \{f(x) : x \in D\}. \quad (1)$$

- $x \in X$: **independent variable**; $y \in R$: **dependent variable**.

Definition 2 (Graph)

If f is a function with domain D , then its **graph** G is the set of ordered pairs (input-output pairs)

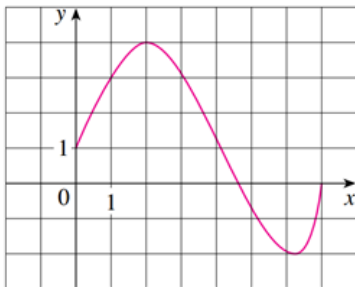
$$G := \{(x, f(x)) : x \in D\}. \quad (2)$$



Example 3

The graph of a function is shown in below Figure.

- (a) Find the values of $f(1)$ and $f(5)$?
- (b) What are the domain and range of f ?



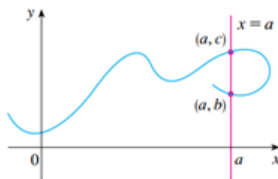
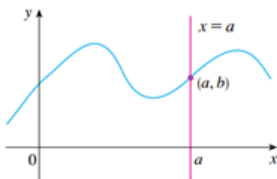
Example 4

Sketch the graph and find the domain and range of each function.

$$(a) f(x) = 2x - 1, \quad (b) f(x) = x^2.$$

The Vertical Line Test

A curve in the xy -plane is the graph of a function of x if and only if no vertical line intersects the curve more than once.



Intercepts

Given a function $y = f(x)$ with domain D_f .

- If $0 \in D_f$ and $f(0) = b$, then the point $(0, b)$ is called the **y-intercept** of the graph of f .
- If $a \in D_f$ such that $f(a) = 0$, then $(a, 0)$ is called the **x-intercept** of f .

Example 5

Find all intercepts of the graph of the function $f(x) = -x^2 + x + 2$.

Representations of functions

There are four possible ways to represent a function:

- Verbally (by a description in words)
- Numerically (by a table of values)
- Visually (by a graph)
- Algebraically (by a formula)

Domain Convention

If a function is given by a formula and the domain is not stated explicitly, the convention is that the domain is the set of all numbers for which the formula makes sense and defines a real number.

$$D_f := \{x \in \mathbb{R} : f(x) \text{ makes sense}\}. \quad (3)$$

Example 6

Find the domain of each function.

$$(a) f(x) = \sqrt{x+2}, \quad (b) f(x) = \frac{1}{x^2 - x}.$$

Example 7 (piecewise-defined function)

A function f is defined by

$$f(x) = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

- 1 Evaluate $f(-2)$, $f(0)$, $f(1)$ and find the domain of f .
- 2 Sketch the graph.

Example 8 (Applying functional notation)

It is known that an object dropped from a height in a vacuum will fall a distance of s ft in t seconds according to the formula

$$s(t) = 16t^2, \quad t \geq 0.$$

- a. How far will the object fall in the first second? In the next 2 seconds?
- b. How far will it fall during the time interval $t = 1$ to $t = 1 + h$ seconds?
- c. What is the average rate of change of distance (in feet per second) during the time $t = 1$ sec to $t = 3$ sec?

Example 9 (Exercise 49, p. 49)

An efficient study of the morning shift at a certain factory indicates that an average worker who arrives on the job at 8:00 A.M. will have assembled

$$f(x) = -x^3 + 6x + 15x^2$$

x hours later ($0 \leq x \leq 8$).

- How many units will such a worker have assembled by 10:00 A.M.?
- How many units will such a worker assemble between 9:00 A.M. and 10:00 A.M.?

Example 10

In a certain state the maximum speed permitted on freeways is 65 mi/h and the minimum speed is 40 mi/h. The fine for violating these limits is \$15 for every mile per hour above the maximum speed or below the minimum speed. Express the amount of the fine F as a function of the driving speed x and graph $F(x)$ for $0 \leq x \leq 100$.

Properties of functions

■ Symmetry

- If a function f satisfies $f(-x) = f(x)$ for every number x in its domain, then f is called an **even function**.
- If a function f satisfies $f(-x) = -f(x)$ for every number x in its domain, then f is called an **odd function**.

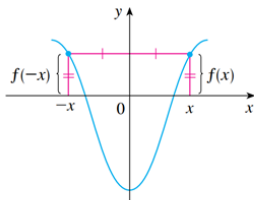


FIGURE 19 An even function

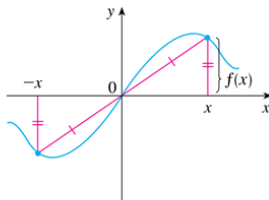


FIGURE 20 An odd function

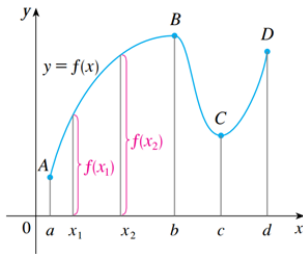
Example 11

Classify the given functions as even, odd, or neither.

$$(a) f(x) = x^2 \quad (b) g(x) = x^3 \quad (c) h(x) = x^2 + 5x.$$

■ Increasing and Decreasing Functions

- A function f is called **increasing** on an interval I if
$$f(x_1) < f(x_2) \quad \text{whenever} \quad x_1 < x_2 \text{ in } I.$$
- A function f is called **decreasing** on an interval I if
$$f(x_1) > f(x_2) \quad \text{whenever} \quad x_1 < x_2 \text{ in } I.$$



Classification of Functions

(1) Polynomial function

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0, \quad (4)$$

where $n \in \mathbb{N}$, $a_0, a_1, \dots, a_n$ are constants called the **coefficients** of the polynomial. If the leading coefficient $a_n \neq 0$, then the integer n is called the **degree** of the polynomial, denoted by $\deg(f) = n$. In particular

- A **constant function**: $f(x) = a$
- A **linear function** is first degree: $f(x) = ax + b$
- A **quadratic function** is second degree: $f(x) = ax^2 + bx + c$
- A **cubic function** is third degree: $f(x) = ax^3 + bx^2 + cx + d$

(2) Rational function

$$f(x) = \frac{p(x)}{q(x)} \quad (5)$$

where $p(x), q(x)$ are polynomial functions.

(3) Power function

$$f(x) = x^a, \quad a = \text{constant}. \quad (6)$$

We consider several cases.

- $a = n$, where n is a positive integer,
- $a = 1/n$, where n is a positive integer,

$$f(x) = x^{1/n} = \sqrt[n]{x}$$

- $a = -1$, the **reciprocal function**

$$f(x) = x^{-1} = \frac{1}{x}$$

with domain $D = \mathbb{R} \setminus \{0\}$.

- (4) **Algebraic function**: A function is called an **algebraic function** if it can be constructed using algebraic operations (such as addition, subtraction, multiplication, division, and taking roots) starting with polynomials

(5) Transcendental function:

- **Trigonometric functions**

$$\sin x, \quad \cos x, \quad \tan x, \quad \cot x, \quad \sec x = \frac{1}{\cos x}, \quad \csc x = \frac{1}{\sin x}.$$

- **Exponential functions:**

$$f(x) = a^x, \text{ where } a \text{ is a positive constant, } a \neq 1.$$

- **Logarithmic functions:**

$$f(x) = \log_a x, \text{ where } a \text{ is a positive constant, } a \neq 1.$$

Equality of functions

Definition 12 (Equality of functions)

Two functions f and g are equal if and only if

- f and g have the same domain.
- $f(x) = g(x)$ for all x in the domain.

Example 13

Given the functions

$$f(x) = 2x - 1, \quad g(x) = 2x - 1, x \neq -3 \quad \text{and} \quad h(x) = \frac{(2x - 1)(x + 3)}{x + 3}.$$

Does $h(x)$ equal $f(x)$ or $g(x)$?

Composition of Functions

Definition 14 (Composite function)

Given two functions f and g , the **composite function** $f \circ g$ (also called the **composition** of f and g) is defined by

$$(f \circ g)(x) = f(g(x))$$

Example 15

If $f(x) = x^2$ and $g(x) = x - 3$, find the composite functions $f \circ g$ and $g \circ f$.

Example 16

Express each of the following functions as the composite of two functions u and g so that $f(x) = g[u(x)]$.

$$(a) f(x) = (x^2 + 5x + 1)^5 \quad (b) f(x) = \cos^3 x \quad (c) f(x) = \sin x^3$$

Example 17 (Exercise 52, p. 50)

At a certain factory, the total cost of manufacturing q units during the daily production run is

$$C(q) = q^2 + q + 900$$

dollars. On a typical workday, the number of units manufactured during the first t hours of a production run can be modeled by the function

$$q(t) = 25t$$

- Express the total manufacturing cost as a function of t .
- How much will have been spent on production by the end of the third hour?
- When will the total manufacturing cost reach \$11,000?

Example 18

A ship is moving at a speed of 30 km/h parallel to a straight shoreline. The ship is 6 km from shore and it passes a lighthouse at noon.

- (a) Express the distance s between the lighthouse and the ship as a function of d , the distance the ship has traveled since noon; that is, find f so that $s = f(d)$.
- (b) Express d as a function of t , the time elapsed since noon; that is, find g so that $d = g(t)$.
- (c) Find $f \circ g$. What does this function represent?

1.5 Inverse Functions, Inverse Trigonometric Functions

Table 1 gives data from an experiment in which a bacteria culture started with 100 bacteria in a limited nutrient medium; the size of the bacteria population was recorded at hourly intervals. The number of bacteria N is a function of the time t : $N = f(t)$.

TABLE 1 N as a function of t

t (hours)	$N = f(t)$ = population at time t
0	100
1	168
2	259
3	358
4	445
5	509
6	550
7	573
8	586

TABLE 2 t as a function of N

N	$t = f^{-1}(N)$ = time to reach N bacteria
100	0
168	1
259	2
358	3
445	4
509	5
550	6
573	7
586	8

Figure: The **inverse function** of f , denoted by f^{-1} , and read “ f inverse.”

Inverse Functions

Definition 19 (Inverse function)

Let f be a function with domain D and range R . Then the function f^{-1} with domain R and range D is the **inverse of f** if

$$f^{-1}[f(x)] = x, \forall x \in D \quad \text{and} \quad f[f^{-1}(y)] = y, \forall y \in R.$$

Example 20

Let $f(x) = 2x - 1$, find f^{-1} , if it exists.

How to find f^{-1} .

- Step 1: Write $y = f(x)$.
- Step 2: Solve for $x = f^{-1}(y)$.
- Step 3: Conclusion $f^{-1}(x)$.

Example 21

If a bacteria population starts with 100 bacteria and doubles every three hours, then the number of bacteria after t hours is $n = f(t) = 100 \cdot 2^{t/3}$.

- (a) Find the inverse of this function and explain its meaning.
- (b) When will the population reach 50,000?

Criteria for Existence of an Inverse f^{-1}

NOTICE: Not all functions possess inverses.

"A function f will have an inverse f^{-1} if it is one-to-one function."

Definition 22 (One-to-one function)

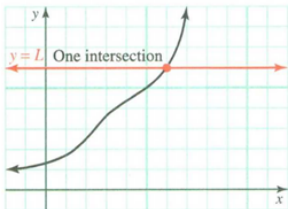
A function f is called a **one-to-one** function if it never takes on the same value twice; that is,

$$f(x_1) \neq f(x_2) \quad \text{whenever} \quad x_1 \neq x_2.$$

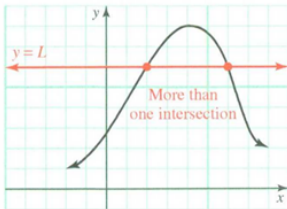
- $f(x) = x^3$ is one-to-one function
- $g(x) = x^2$ is not one-to-one.

Horizontal Line Test

A function is one-to-one if and only if no horizontal line intersects its graph more than once



a. A function that has an inverse



b. A function that does not have an inverse

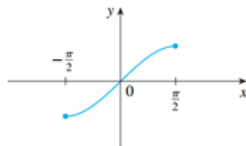
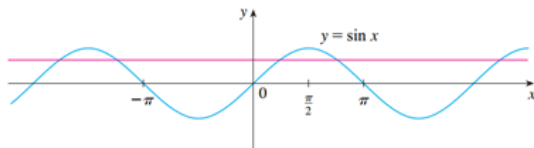
CAUTION: Do not mistake the -1 in f^{-1} for an exponent. Thus

$$f^{-1}(x) \text{ does not mean } \frac{1}{f(x)}.$$

The reciprocal $1/f(x)$ could, however, be written as $[f(x)]^{-1}$.

Inverse Trigonometric Functions

■ Sine function: $y = \sin x$



The **inverse sine function** $y = \sin^{-1} x$ if

$$\sin y = x \quad \text{and} \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

The cancellation equations are

$$\sin^{-1}(\sin x) = x \quad \text{for} \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2},$$

$$\sin(\sin^{-1} y) = y \quad \text{for} \quad -1 \leq y \leq 1.$$

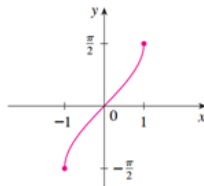


Figure: $y = \sin^{-1} x = \arcsin x$

Inverse Trigonometric Functions

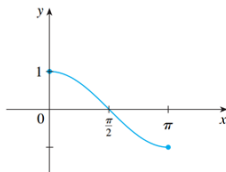


Figure: $y = \cos x$

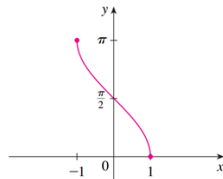


Figure: $y = \cos^{-1} x$

The **inverse cosine function** $y = \cos^{-1} x = \arccos x$ if

$$\cos y = x \quad \text{and} \quad 0 \leq y \leq \pi$$

The cancellation equations are

$$\cos^{-1}(\cos x) = x \quad \text{for} \quad 0 \leq x \leq \pi,$$

$$\cos(\cos^{-1} y) = y \quad \text{for} \quad -1 \leq y \leq 1.$$

Inverse Trigonometric Functions

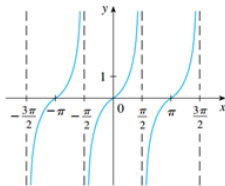


Figure: $y = \tan x$

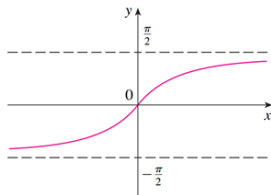


Figure: $y = \tan^{-1} x = \arctan x$

The **inverse tangent function** $y = \tan^{-1} x$ if

$$\tan y = x \quad \text{and} \quad -\frac{\pi}{2} < y < \frac{\pi}{2}$$

The cancellation equations are

$$\tan^{-1}(\tan x) = x \quad \text{for } -\frac{\pi}{2} < x < \frac{\pi}{2},$$

$$\tan(\tan^{-1} y) = y \quad \text{for all } y.$$

Example 23

Evaluate the given functions

a. $\sin(\sin^{-1} 0.5)$

b. $\sin(\sin^{-1} 2)$

Example 24 (Exercise 53, p. 60)

A painting 3 ft high is hung on a wall in such a way that its lower edge is 7 ft above the floor. An observer whose eyes are 5 ft above the floor stands x ft away from the wall. Express the angle θ subtended by the painting as a function of x .

Example 25 (Exercise 54, p. 60)

To determine the height of the building, select a point P and find the angle of elevation to be α . Then move out a distance of x units (on a level plane) to point Q and that the angle of elevation is now β . Find the height h of the building, as a function of x .

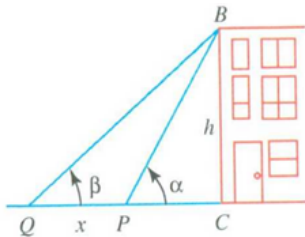


Figure: Determining height

Mathematical Models

A **mathematical model** is a mathematical description (often by means of a function or an equation) of a real-world phenomenon such as

- the size of a population
- the demand for a product
- the speed of a falling object, etc.

The purpose of the model is to understand the phenomenon and perhaps to make predictions about future behavior.

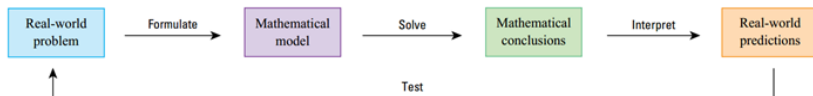


FIGURE 1 The modeling process

Example 26

Table 1 lists the average carbon dioxide level in the atmosphere, measured in parts per million at Mauna Loa Observatory from 1980 to 2008. Use the data in Table 1 to find a model for the carbon dioxide level.

TABLE 1

Year	CO ₂ level (in ppm)	Year	CO ₂ level (in ppm)
1980	338.7	1996	362.4
1982	341.2	1998	366.5
1984	344.4	2000	369.4
1986	347.2	2002	373.2
1988	351.5	2004	377.5
1990	354.2	2006	381.9
1992	356.3	2008	385.6
1994	358.6		